

Avishek Dutta

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RESEARCH INTERESTS

Applied artificial intelligence and machine learning, natural language processing for low-resource languages (Bengali), machine learning for biomedical signal analysis, digital governance and technology adoption, human–computer interaction, and institutional automation systems.

EDUCATION

Bachelor of Science in Computer Science and Engineering

May 2025 – Present

*Department of Electrical and Computer Engineering
North South University, Dhaka, Bangladesh*

Higher Secondary Certificate, Science

February 2023 – November 2024

Cantonment College, Cumilla Cantonment, Bangladesh

Secondary School Certificate, Science

January 2017 – October 2022

Feni Government Pilot High School, Feni, Bangladesh

SKILLS & TOOLS

Programming Languages

Python, TypeScript, C/C++, Java, Rust, Bash, PowerShell, SQL, Assembly

Frameworks and Libraries

React, Next.js, Express.js, FastAPI, Flask, Node.js, LangChain, Tailwind CSS, React Native, NumPy, Pandas, Scikit-learn, Seaborn, Keras

Databases and Infrastructure

PostgreSQL, REST APIs, JWT, Google OAuth, CI/CD, Docker, Cloud Deployment

Security

CSRF Protection, Input Sanitization, Rate Limiting, Password Hashing, System Design, Secure Authentication Design

Tools and Platforms

Git/GitHub, LaTeX, Linux, Figma, Adobe Illustrator

ACADEMIC AND PROFESSIONAL APPOINTMENTS

Student Worker, Office of Student Affairs

May 2026 – Present

North South University, Dhaka, Bangladesh

Provide administrative and operational support within the Office of Student Affairs, including institutional record-keeping, documentation, and data management. Maintain digital archives for accu-

racy and accessibility, assist with website and digital content updates, and support coordination of co-curricular and extracurricular activities.

In-Charge

October 2025 – Present

NSU Computer & Engineering Club, Dhaka, Bangladesh

Lead technical operations, launching scalable platforms like GAMEWAVE 2.0 (1,000+ participants) and Cyberantus 2026 (5,000+ participants, 1,000+ teams). Additionally, drive Corporate & Outreach by preparing proposals and negotiating partnerships, while supporting operations across Admin & HR, PR, and Event Logistics.

Platform Manager

May 2025 – Present

CSE4ALL, Dhaka, Bangladesh

Manage platform operations for an initiative delivering Computer Science education without formal prerequisites. Oversee a curriculum combining self-paced video instruction, live sessions, assignments, and certification, aimed at learners across Bangladesh.

Executive Designer

November 2021 – March 2022

Mind Splash Magazine, Dhaka, Bangladesh

Designed visual identity and feature layouts for print and digital publication; collaborated with editorial teams on content presentation.

SELECTED PROJECTS

Arrhythmia Detection System

Trained a convolutional neural network on the MIT-BIH Arrhythmia Database to classify cardiac arrhythmias, handling data loading and preprocessing with NumPy and pandas and model training with Keras, achieving 98.65% classification accuracy. Integrated the trained model into a mobile application via a REST API, enabling real-time inference within a React Native (Expo) frontend.

Python, NumPy, Pandas, Keras, Flask, React Native (Expo)

NSU Student Affairs Internal Management System

Developed an internal platform for the Office of Student Affairs to digitize and streamline administrative workflows, including document and paper tracking, structured volunteer data submission from student clubs, automated ID card generation, and automated document generation in document format.

Next.js, PostgreSQL, System Design & Automation, RBAC

NSU CEC Cybernauts 2026 Website and Management System

Developed a high-performance frontend for the official Cybernauts 2026 platform, handling registration and coordination for a national-level event with asynchronous processing, zero downtime, and near-zero infrastructure investment. Implemented a multi-role admin panel for event management, secure authentication via Google OAuth, dynamic custom registration forms, and layered security measures for data integrity.

Next.js, TypeScript, PostgreSQL, Google OAuth

cybernauts.nsucec.com

NSU CEC GAMEWAVE 2.0 Platform

Independently built a full-stack event platform for North South University's flagship gaming event, including frontend, backend, registration system, admin panel, and DevOps deployment, engineered to sustain high traffic during peak participation.

TypeScript, React.js, Express.js, Tailwind CSS

gamewave.nsucec.com

RoktoJog — End-to-End Blood Donation System

Designed the system architecture for a blood donation and management platform, encompassing donor registration, blood inventory tracking, appointment scheduling, and emergency request handling with secure authentication.

System Architecture, JWT, Password Hashing

NSU CEC Website and Management System

Built the official website and management system for the NSU Computer & Engineering Club, featuring event management, member registration, and role-based access control for administrators and members.

nsucec.com

Everyday One Taka Fund — Micro-Donation Platform

Developed a member management system for a micro-donation platform, integrating secure payment gateway processing, automated donation tracking, and a member dashboard with contribution history.

everyday1takafund.com

BanglaCode — C Programming in Bangla Syntax

Built an experimental programming language mimicking C syntax using Bangla characters, including a custom lexer, parser, and runtime environment, demonstrating language design and interpreter construction.

Interpreter Design, Computer Programming

banglancode.pages.dev

CPawnX — Bitboard-Based Chess Engine

Built a chess engine from scratch in TypeScript using a magic bitboard architecture for move generation, without reliance on external chess libraries or frameworks. Verified move-generation correctness through perft testing across multiple search depths.

TypeScript, Bitboard Representation, Move Generation, Next.js

cpawnx-engine.vercel.app

Basic System Kernel (BSK)

Developed a basic operating system kernel using Assembly and C to demonstrate fundamental OS concepts, including bootloader implementation, memory management, interrupt handling, and low-level I/O.

Assembly, C, Systems Programming, System Architecture

gitlab.com/tyroruyk/bsk

HONORS AND AWARDS

Champion, Future Builders: AI-Driven Cognitive Innovation Hackathon (IEEE)

2025

Led team NSU_MissionX in developing *ShasthoBondhu*, an AI-powered health companion, under competitive time constraints.

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

Working with Hugging Face

DataCamp, Jul 2026

Credential ID [e533684e131c658013d9fed9ecb9a444c47d838d](#)

Prompt Engineering with the OpenAI API

DataCamp, Jun 2026

Credential ID [bb2f63759a8c13d3f5ffe3a40844cb70ae23cc54](#)

Virtual Youth Summit for Human Rights 2026

University of Arizona, College of Social &

Behavioral Sciences, Mar 2026

Certificate of Participation

IEEE Student Member

IEEE, Oct 2025

Credential ID 101425164

Certificate of Appreciation — 3rd National ICT Scout Jamboree 2025

Bangladesh Scouts,

Sep 2025

Credential ID TCS0597/3

AI Prompt Engineer Level 1™

AI CERTs®, Jul 2025 – Jul 2026

Credential ID 28a3e8cde967

Open Researcher and Contributor ID (ORCID)

ORCID, Feb 2024

Credential ID: 0000-0002-8328-8649

Ready Player 50 (CS50)

Harvard University, Oct 2023

Credential ID bb77190a-dfdd-4884-9ecf-b5eef9ec1de9

OUTREACH AND PUBLIC ENGAGEMENT

Contributing Writer

2022 – 2025

Banger Chatar Biggan

Author science communication articles in Bengali on physics, mathematics, computer science, and emerging technologies, translating research in AI, quantum computing, and computational science for general audiences.

Training and Development Executive

2020 – 2022

FGPHS ICT Club, Feni, Bangladesh

Designed and delivered training curricula in programming, web development, and cybersecurity fundamentals; mentored over 100 students through workshops and hands-on projects.

LANGUAGES

Bengali – Native proficiency | **English** – Professional proficiency

Last updated: July 10, 2026